Rigour Without Infrastructure

Three Propositions on Claim-Card Discipline as a Substitute for Institutional Certification

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In brief. Problem. A person with no university, no lab, and no research team who now has AI assistance faces a new failure mode: text that reads as smart and academic while it stays unclear which part is evidence, which part is interpretation, and which part AI smoothed over past what the data supports (Section 1). **Proposition.** Three candidate hypotheses (Section 5) — a design mechanism (H1), a conceptual claim about where rigour lives (H2), and an empirical claim about cross-vendor AI checking (H3) — propose that a claim-card discipline bound to Existence–Attribution–Disclosure and gated by an independence ladder could make a standalone scholar's claim reader-checkable without an institutional credential. **Method.** Each proposition carries a named falsifier and is placed against 48 independently verified citation cards in a conversation-with-the-literature table (Section 6), then the whole method is run on this paper's own construction as a self-application demonstration (Section 7), including an independent review of the three claim cards themselves by three cross-vendor AI routes, which revised them (Section 7). **Status.** All three propositions are carried as untested claims at tier DR, none above knowledge state K0; H3 is explicitly untested, and the self-application run itself surfaced real gate failures, not a clean pass (Section 7, Section 8).

Abstract. People who live daily with a problem often know it before researchers do, but turning that knowledge into a checkable claim has historically required a university, a lab, or a research team. Generative AI removes that requirement technically while introducing a new one epistemically: it becomes cheap to produce text that reads as rigorous without it being clear which part is observation, which part is inference, and which part the AI supplied. This paper states, as three falsifiable propositions and not as demonstrated results, a candidate answer: a *claim card* that separately records what was seen, what the data separates, what AI filled in, what is assumed, and what independent check (if any) the claim has survived — bound to a formal Existence–Attribution–Disclosure norm and passed through a named independence ladder (I0 self-check through I5 outside human review) before public release — could carry the checkable property institutions have historically supplied, without requiring the institution itself (H1, design; H2, conceptual). A third, narrower proposition (H3, empirical) holds, untested, that cross-vendor AI checking would measurably outperform same-vendor re-checking at catching undisclosed AI-fill; no measurement of that difference has been made (n=0). Each proposition is placed against a literature conversation built from 48 independently verified citation cards, and the whole method is applied reflexively to this paper's own construction, which surfaced real failures — a validator warning, a manifest-detected search-mode gap, and a documented citation-verification correction — reported as findings, not smoothed into a clean success story. The three claim cards carrying these propositions were themselves reviewed by three cross-vendor AI routes and revised in response before this draft (Section 7). Every claim here sits at tier DR and knowledge state K0; none has passed an independent (I5) human check.

Positioning. We are not competing with knowledge-authority and make no priority claim; we state what we propose, what we build on, and what would make us wrong.

1 PROBLEM BEFORE OBSERVATION

"Observation" is abstract; it is a verb, not a starting node. Founder's line, English gloss below (Blackbox Log `BBL-2026-09-04-083`, concept DOI `10.5281/zenodo.22302518`):

> *Founder's line (English gloss; verbatim Thai in the Blackbox Log, DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-083`):*
> — "the word 'observation' is abstract, it is a verb; if we started from the *problem* instead, is there a process like this? We propose focusing on the problem first."

A problematic state exists in the world *before* any knower acts on it — before it is looked at, named, or measured. The spine this paper's claims are built on (adopted this pass into `design/FOUNDATION_v0.5.md` §2.1a, BBL-2026-09-04-084) therefore does not start with "Reality →" as its first node — the lens never grants direct access to reality, only to what was read out ($R_A = O_A(W;\Pi_A) \neq W$) — it starts one layer finer:

> Problem state (unread) → first human readout (the Blackbox Note's raw line, verbatim, untranslated) → human-language question → [lens-in] → hypothesis (lens-out, signed) → evidence → provisional knowledge (K0, ...)

This paper's own problem state, before it was read out at all: a person who lives daily with a problem, has no university appointment, no certified lab, and no standing research team, and now has general-purpose AI assistance. The first human readout of that state (Thai, source of truth), Blackbox Note line 1, English gloss below (`blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md#L1`):

> *Founder's line (English gloss; verbatim Thai in `blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md#L1`):*
>
> — "how can a person with no university, no lab, and no research team produce rigorous knowledge from their own site of practice?"

The contaminated concept this question already flags: "rigour" ordinarily presupposes an institutional mechanism (a lab, a department, a review committee) sitting behind a claim, rather than naming a property a single claim can carry on its own, field by field.

2 Standpoint and ownership

This paper is written from the standpoint of a social-enterprise practitioner and citizen (the founder, Yaoharee Lahtee), not a substitute expert in computer science, philosophy of science, library/information science, or research-integrity studies. **Disciplines not claimed:** medicine, religious/fiqh authority, law, engineering, anthropology, political science, and formal computer science beyond what is mechanically executed here. This is disclaimer **D-STANDPOINT / D-NONEXPERT** in Section 8.

2.1 Responsibility per arrow

Every arrow on the spine above is named by who performed it. **Data** $\rightarrow$ **Inference** may be `human`, `ai`, or `joint`; the three project claim cards behind this paper (Section A) record it as `joint` — AI retrieved literature and drafted candidate hypotheses, the founder framed the problem and question. **Inference** $\rightarrow$ **Claim is always human.** AI is never an author. This is not a new rule so much as this pass naming, per arrow, a distinction the repo already held at the whole-artifact level (`AGENTS.md` gate rule 5).

2.2 Ownership: problem, question, and the selection of the hypothesis stay human

Founder's line, English gloss, first stated (`BBL-2026-09-04-086`) and then self-corrected (`BBL-2026-09-04-088`, the correction shown here is the one adopted):

> *Founder's line (English gloss; verbatim Thai in the Blackbox Log, DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-088`):*
>
> — "if the problem is still ours, the question is still ours, and the *selection* of the hypothesis is still ours, and it leads to solving our problem, then use AI, use a cat, use a bear, go ahead — because that is the goal of knowledge: to solve a problem for someone."

Tag	Meaning
TH_COQC	Machine-checked, axiom-free formal result.
FINITE_DIAGNOSTIC	Measured/computed on a finite, retained, reproducible input; not a limit or population claim.
FIT_CALIBRATED	A model calibrated against declared data; accuracy conditional on that fit.
DR	Declared bridge / human narrative synthesis; a design or interpretive judgment, not a proof or measurement.
DEFINITION	A stipulated definition used only where decision-bearing.
OPEN	Not established; a real open question, not smoothed over.

Table 1: Claim-tier legend (readout_genesis + readout_universe ladder). Every propositional claim in this paper (H1, H2, H3, Section 5) is DR; the demonstration (Section 7) is FINITE_DIAGNOSTIC where a command was actually executed.

Three acts on the spine are the human's: owning the **problem**, asking the **question**, and **selecting** the hypothesis. Every other node — retrieval, drafting, analysis, checking, rendering — may be performed by any tool. Concretely for this paper: AI (the AI assistant) drafted candidate hypotheses H1, H2, and H3 (`hypotheses.md`); the founder's own act of selection, recorded in `hypothesis_selection.yaml` and confirmed by ruling `BBL-2026-09-04-095` (Thai verbatim in `entries.jsonl`; gloss: "bring all three together"), chose all three — H1 and H3 are carried as untested propositions with stated falsifiers, and H2 is advanced in the paper's own voice as its central conceptual claim (Section 4, `genre_decision.yaml`).

3 The lens and the five questions

3.1 Claim-tier legend

Every claim in this paper carries exactly one tier tag; tiers are never collapsed or silently upgraded (Table 1).

3.2 The readout-not-truth lens

This paper's problem-to-hypothesis translation was performed under *Readout Universe — Yaoharee Lahtee* (lens; DOI 10.5281/zenodo.21529456, 10.5281/zenodo.21665100; repositories `https://github.com/morrocwi/readout_universe`, `https://github.com/morrocwi/readout_genesis`). Naming the lens credits its source and lets a reader replace it; it does not make the lens, or any hypothesis drafted under it, true (disclaimer **D-EXTERNAL-INPUT**). Everything read under this lens is a finite, retained *readout* of a source, never the source itself: a benchmark number, a citation's exact passage, a claim card's own field, this paper's own prior draft — each was produced by some process, at some resolution, under some method, and is read as that readout, not as truth handed down.

3.3 Existence–Attribution–Disclosure and the five questions

The Readout Condition is $E \wedge A \wedge D$: a claim satisfying Existence (some provenance path for every load-bearing distinc-

#	Founder's question	Mechanically check-able?
Q1	what was actually seen	presence only
Q2	what the data separates	presence only
Q3	what AI filled in	presence only
Q4	what is assumed	presence only
Q5	what independent evidence/objection	**the one genuinely enforce-able check**

Table 2: Founder's five questions ⇔ E-A-D ⇔ claim-card field group (`design/FOUNDATION_v0.5.md` §3.1). A card that answers Q1/Q2/Q4 honestly but misattributes Q3 — crediting the source with a distinction AI actually supplied — fails the Readout Condition even though every field looks filled; `silent_lift_check` is the mechanized test for exactly this. Thai originals: Blackbox Log `BBL-2026-09-04-083` ff.

tion), Attribution (identification of every non-source node), and Disclosure (every AI-supplied distinction named, not silently absorbed into the claim) is what a claim card is built to record, field by field, against the founder's own five questions (Table 2).

Only Q5 — whether an independent check was ever run, and at what independence class — is structurally enforceable by the kernel; Q1–Q4 are presence-checkable, never correctness-checkable, because *mechanical validity ≠ semantic validity*. This is why the independence ladder (Section 6, Section 7) is the licensed route to closing what the other four questions cannot mechanically close on their own.

4 Genre and evidence standard

This paper is genre-routed as **CONCEPTUAL** by a human, adopted-by-human override of the router's own mechanical output (`genre_decision.yaml`). The router (run with `-search-log`) returned `systematic_review` because a frozen `search_log.yaml` exists under each of `h1/h2/h3`; the founder's ruling (`BBL-2026-09-04-097`, Thai verbatim in `entries.jsonl`; gloss: "make this document an academic article, a concept paper") does not adopt that output — a documented search log supporting three candidate hypotheses is not the same fact as the genre of the claim the paper itself advances, and the router has no way to see that distinction (`genre_decision.yaml` note). H2 (the conceptual, portable-property claim) is advanced in the paper's own voice; H1 (design) and H3 (empirical) are carried as propositions with stated falsifiers, neither run as a study this pass. Evidence standard: every citation is a `status: VERIFIED` card (Section 6); every propositional claim stays DR until an independent test says otherwise.

5 Three propositions

Candidates were drafted by AI from the lens translation above; the founder selected among them (Section 2). All three are carried as propositions, not results.

H1 — Design: the claim-card mechanism ([C1])

Statement. If every claim a standalone scholar produces is recorded as a claim card answering the five questions, tied to E-A-D, and passed through the independence ladder before any public release, then a person with no institutional affiliation *could, as a design proposition,* co-produce knowledge with AI whose rigour a reader can check for themselves, without an institutional credential serving as the certifying step. This is a proposed mechanism, not an established one: this project's own single self-application (n=1, this paper's own production run, Section 7) is offered as one demonstration readout of the mechanism, not as a general test across standalone scholars.

Falsifier. A claim card that passes every field the kernel can mechanically check is nonetheless found by a genuine independent reviewer (I5) to have a false answer at Q2 or Q3 — showing the mechanical check does not track the epistemic property it claims to track. Not yet run: this project has not secured an I5 (outside human, no stake) route, so this falsifier has not been tested in either direction.

What would count as evidence. A corpus of claim cards run through the independence ladder to at least I3, with a subset reaching genuine I5, compared against a matched corpus produced without the discipline, scored on whether an independent reader can recover the evidence/AI-fill boundary the maker recorded.

H2 — Conceptual: rigour as a portable claim-level property ([C2])

Statement. "Rigour" is not a property that institutional machinery (lab, department, review committee) confers on a claim from the outside; it is proposed here, as a conceptual contrast, to be *at least in part* a property a single claim can carry in its own recorded structure — institutions are one historical mechanism that has produced this property, not shown here to be the only possible one. This does not establish that institutions are non-load-bearing: the cited evidence for it is mixed and mostly same-lineage (Table 4).

Falsifier. A claim card that answers all five questions honestly and completely, checked at I3, is shown to still require an institutional credential — not just an independent human reviewer — to be treated as rigorous by its actual intended readers, for at least one class of reader/decision this project cares about.

What would count as evidence. Reader studies on whether a target audience (journal editors, grant reviewers, a practitioner's own community) treats a well-formed claim card as sufficient without also asking who backs it.

H3 — Empirical: cross-vendor checking raises AI-fill detection ([C3], untested)

Statement. Hypothesis, not yet tested: when claim cards are checked by a different AI vendor from the one that produced them (cross-vendor, independence level I3) instead of being re-checked by the same vendor that produced the card (I0/I1), the rate at which undisclosed AI-fill (a false Q3 field) or undisclosed assumptions (a false Q4 field) is caught *would rise* measurably compared to same-vendor re-checking. No seeded-error benchmark has been run by this project (n=0 own trials); the single most directly on-point source found the effect direction-dependent, not uniform (Table 5).

Falsifier. A controlled comparison (claim cards seeded with a known number of undisclosed AI-fill/assumption instances, same-vendor vs. cross-vendor routes) shows no significant difference, or shows cross-vendor performs worse.

What would count as evidence. A seeded-error benchmark with a pre-registered scoring rubric; tiered FIT_CALIBRATED at best until run, DR until then. **This paper does not run that benchmark; H3 is stated, not tested.**

All three propositions were drafted under the same lens (Section 3); none has an evidence relation above independence class I3, and none of the underlying claim cards has yet been approved by the founder for release (`independent_check.status: PENDING` on all three — checked by cross-vendor routes R1–R3 and revised accordingly, not yet re-checked by a new route or founder-approved, Section A).

6 CONVERSATION WITH THE LITERATURE

Each table below is a *conversation with the problem*, not a chronology or priority ranking (`design/S14_literature-review-system.md`): sources are placed by how they talk to the proposition, never by who came first. Every row resolves to a citation card with `status: VERIFIED` in the corresponding `litreview_manifest.yaml` (`records/lit/rigour-without-infrastructure/h{1,2,3}/`), meaning both `metadata_verified` (Crossref/OpenAlex/arXiv lookup) and `claim_match_verified` (a decorrelated cross-vendor AI route, `route:cross-vendor-ai`, I3) hold. **Relation column** follows the repo's comparison rule (same/different/cited, never a priority claim): *same* = the source's own finding is compatible with the proposition's direction; *different* = the source's own finding counts against it; *cited* = the source bears on the topic without a same/different verdict (orthogonal or undetermined in the underlying dialogue table); rows marked *own lineage, not independent* are the founder's own prior published pieces, collided against world literature per instruction `BBL-2026-09-04-098` and never counted as independent support.

Across all three tables, 20 of 48 rows (42%) are the founder's own prior published pieces, collided against world literature per instruction `BBL-2026-09-04-098` rather than left silent; every own-lineage row is marked *own lineage, not independent* and none is counted toward any of the three propositions' evidential support.

7 THE ARTIFACT APPLIED TO ITSELF

This section reports what the 2026-09-04 self-application run — glosa run on glosa's own paper, per instruction `BBL-2026-09-04-091` — actually recorded, as a readout of that run, not as a claim that the method passed cleanly.

Gate results actually executed

Running `./cli/glosa claim validate` directly against the three project claim cards (`GLOSA-CC-20260904-0201/0202/0203`) this session returns `verdict: PASS_WITH_LIMITS` for all three, with zero schema errors, but with one identical warning on every card: *"D-LENS-UNCITED not checked – validate_claim_card was not given citation_cards (lens_translation.lens_ref is set); pass citation_cards=[. . .] to close this gap."* The validator is honest about its own coverage limit here: a schema-level PASS is not, by itself, a claim that the lens-citation gate closed — it closed only once this session additionally cross-checked the lens DOIs against the citation-card apparatus described in Section 3.

Composite quotes caught by the cross-vendor route

The single strongest gate finding this run: an I3 (decorrelated cross-vendor) check on an earlier citation pass caught *composite quotes* — citations assembled from memory rather than a locator taken from the opened source. The founder's ruling in direct response, English gloss (`BBL-2026-09-04-100`):

> *Founder's line (English gloss; verbatim Thai in the Blackbox Log, DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-100`):*
>
> — "should we make it a hard rule: if a source is used it must carry a page and line locator plus a full link, never written from memory — that reduces the problem at the source, before the check."

Adopted immediately (`BBL-2026-09-04-101`, Thai verbatim in `entries.jsonl`; gloss: "update it into the system"), this became the rule under which every one of the 48 citation cards behind Section 6 was subsequently built: each carries `page_or_locator` and `fetched_from_url`, and each `exact_passage` is a source-first quotation, not a recalled paraphrase. This is a real, gate-caught failure that changed the method's own rule (rule 17, `design/FOUNDATION_v0.5.md` §7.8), reported here because it happened, not because it flatters the method.

Citation card	Source	Relation	Bearing
cite-cstp-boundary-work-001	Mayes 2022, Citizen Science and Scientific Authority	different	Moderate, scoped: credentialed-expert legitimacy-shaping work continues to matter in practice — cross-vendor review narrowed this to bear only on credentialed legitimacy/acceptance pressure, not to be read as evidence the full claim-card + E-A-D + independence-ladder mechanism was tested and found wanting.
cite-disclosure-not-documentation-001	Cwik 2026, Disclosure is not documentation	same	Weak, component-only: supports the H1 component that AI-assisted work needs structured documentation beyond disclosure — cross-vendor review downgraded this from moderate to weak and scoped it to `SUPPORTS_COMPONENT_ONLY`, not evidence the combined mechanism was tested.
cite-llm-hallucination-survey-001	LLM hallucination survey, arXiv:2510.06265	cited	Shows AI cannot be trusted to flag its own errors — exactly why H1 requires an external check rather than AI self-report.
cite-ocsdnet-honf-001	OCSDNET/HONF, Open Hardware and Citizen Science (Indonesia, Thailand, Nepal)	same	Grounds H1's premise that non-institutional knowledge production is a live regional practice, not a hypothetical.
cite-own-blackbox-log-h1-006	Blackbox Log (Lahtee, Zenodo)	own lineage, not independent	Documents the append-only raw-record practice H1 assumes is followed; no claim about rigour by itself.
cite-own-readout-condition-h1-002	The Readout Condition (Lahtee)	own lineage, not independent	Supplies the E-A-D norm H1's claim card operationalizes; does not itself specify the card's fields.
cite-own-readout-genesis-h1-003	Readout Genesis Standalone Synthesis (Lahtee)	own lineage, not independent	Supplies the independence-ladder vocabulary H1 tracks; does not itself define a claim-card schema.
cite-own-standalone-scholar-h1-005	The Standalone Scholar (Lahtee)	own lineage, not independent	Names the dual-track conversion (AI-assisted formation → independent human friction) H1's card is scoped inside, not a replacement for.
cite-own-written-ai-h1-001	Written by AI. Still True. (Lahtee)	own lineage, not independent	Names "knower fetishism"; H1's card targets exactly the constitution-level property this paper argues authority derives from.
cite-own-zero-readout-h1-004	What a Zero Readout Certifies (Lahtee)	own lineage, not independent	Illustrates machine-checked discipline; H1's own mechanical checks (schema validation) are markedly weaker.
cite-pgph-citizen-science-sea-001	Kumar et al. 2025, Community perceptions of citizen science (South/SEA)	cited	Governance-inclusion framing, orthogonal to H1's rigour-checkability framing.
cite-rsos-citizen-epistemology-001	Jaeger et al. 2023, An epistemology for democratic citizen science	same	Weak, general-context only: robust knowledge can arise through plural/citizen epistemology outside narrow institutional credentialing — cross-vendor review confirmed this as `SUPPORTS_GENERAL_CONTEXT` background support, not upgraded to a test of H1's mechanism.
cite-rsos-preprint-credibility-001	Soderberg, Errington & Nosek 2020, Credibility of preprints	same	Researchers rate content/process cues over prestige when prestige is absent — supports the premise, not H1's specific design.
cite-scholarly-kitchen-credentialism-001	Scholarly Kitchen 2026, The Guild and the Gap	cited	A real rejection episode on prior-publication grounds; whether a claim card would change the outcome is untested.
cite-tcij-freelance-th-001	TCIJ (TH) 2020, freelance-academic labour piece	cited	Adjacent labour/credit question, thinner on H1's actual rigour-verification topic than its search match suggested.

Table 3: H1 (design proposition) against 15 verified citation cards, 6 of them own-lineage. Diversity audit (`h1/litreview_manifest.yaml`): 1 Thai / 5 English external sources (own-lineage excluded), stance counts *agrees*:4 *orthogonal*:2 *undetermined*:9 by the manifest's own collapsing convention (Section 7 notes a discrepancy between this convention and the citation cards' own explicit notes). `concentration_flags: []`; `zero_disagree_flag: false`.

A manifest-detected search-mode gap

The genre router's own mechanical output (Section 4) returned `systematic_review` because `context.has_search_log` was true — but every one of `h1/h2/h3`'s own `litreview_manifest.yaml` records `search_mode` as `TARGETED_SEARCH / SCOPING_SEARCH`, never `SYSTEMATIC_REVIEW` — the manifest, read directly, is what let the founder catch that the router's mechanical genre suggestion did not match the actual evidentiary standard the literature runs had met, and overrule it (Section 4). This is the manifest doing exactly the job `design/FOUNDATION_v0.5.md` §7.8's **FC-S8-1** hard block exists for: calling anything less than a full SR protocol a "systematic review" is blocked, here by the human reading the manifest against the router's own claim.

A stance-counting discrepancy this session found and did not paper over

Building Tables 3 and 5 directly from each citation card's own `notes` field ("Dialogue-table stance: NO (disagrees with / challenges H1)" on `cite-cstp-boundary-work-001`) surfaced a discrepancy with the corresponding manifest's own `diversity_audit.counts.by_stance`: the manifests for H1 and H3 both record `disagrees: 0`, while this session confirmed directly (grep against the raw dialogue-table rows, FINITE_DIAGNOSTIC) that at least one row per hypothesis (Mayes 2022 against H1; Xiang et al. 2026 against H3) is an explicit, single-value challenge row that a card's own notes field independently labels a disagreement. The manifest's aggregate counting convention appears to collapse a dual-value "NO/NO" dialogue-table cell into `undetermined` rather than `disagrees`, undercounting real challenges in its own summary statistic by at least one row per hypothesis where this occurs. This paper's own tables (Tables 3 and 5) use the citation card's explicit `notes` field, not the manifest's aggregate count, and this discrepancy is disclosed rather than silently reconciled — a small instance of exactly the "mechanical check does not track the epistemic property it claims to track" failure mode H1's own falsifier names, found here in the tool that mechanically summarizes H1's own supporting literature.

Path leaks caught by CI, not by this run

The repository's public-facing leak gate (`scripts/check_leak.sh`, wired into `.github/workflows/ci.yml`'s `leak-scan` job) is a

Citation card	Source	Relation	Bearing
cite-rigour-without-infrastructure-001	Wikipedia, Open-notebook science	cited	Treats openness as a property of record-keeping practice, close to but coarser than H2's own five-field structure.
cite-rigour-without-infrastructure-002	Rasmussen 2019, Confronting Research Misconduct in Citizen Science	cited	Calls for institution-like misconduct-detection machinery — cuts against H2's claim that the property needs no such mechanism.
cite-rigour-without-infrastructure-003	Freitag, Meyer & Whiteman 2016, credibility-building strategies	same	Credibility built bottom-up through concrete recorded practices, not top-down certification.
cite-rigour-without-infrastructure-004	Scholarly Kitchen 2026, The Guild and the Gap (+ comments)	different	Close to H2's own falsifier: a named commenter states affiliation/publication history earns *more* scrutiny than any single record could substitute for.
cite-rigour-without-infrastructure-005	Dutta et al. 2021, Decolonizing Open Science	cited	Structural/platform-governance framing, adjacent to but different from H2's single-claim scope.
cite-rigour-without-infrastructure-006	Okafor et al. 2022, Institutionalizing Open Science in Africa	cited	Aims for *more* institutional machinery as the path forward — the opposite emphasis from H2, at a different level.
cite-rigour-without-infrastructure-007	Kumar et al. 2025, community perceptions of citizen science	cited	Access to institutional audiences, not record quality, is the bottleneck this source finds — adjacent friction H2 does not resolve.
cite-rigour-without-infrastructure-008	Srirot & Sohsomboon 2025 (TH), qualitative Trustworthiness framework	same	Record-level techniques (audit trail, triangulation) let a reader judge rigour without institutional standing — the closest structural match to H2 found.
cite-rigour-without-infrastructure-009	THE STANDARD (TH) 2025, Thai Open Science policy piece	cited	Institution-led opening of an existing record — the institution still decides what counts as legitimate, an access question not a rigour-provenance one.
cite-own-blackbox-log-h2-006	Blackbox Log (Lahtee)	own lineage, not independent	No claim about whether rigour is portable at claim level.
cite-own-civilization-of-knowledge-h2-007	The Civilization of Knowledge (Lahtee)	own lineage, not independent	Locates the stabilizing step historically in institutions — would ask whether a claim card is a new non-institutional stabilizer or a mistaken substitute.
cite-own-readout-condition-h2-002	The Readout Condition (Lahtee)	own lineage, not independent	Weak, indirect: supports source-relative warrant architecture, not explicitly "rigour" or institutional non-conferment — cross-vendor review downgraded this from moderate/direct to weak/indirect support; own lineage, not independent confirmation.
cite-own-readout-genesis-h2-003	Readout Genesis Standalone Synthesis (Lahtee)	own lineage, not independent	Institutions are one belief-scale among several, not a precondition for knowledge status.
cite-own-standalone-scholar-h2-005	The Standalone Scholar (Lahtee)	own lineage, not independent	Independent human friction is one specific historical route to stress-tested knowledge, not necessarily the only one — would ask if H2 still needs some friction step.
cite-own-written-ai-h2-001	Written by AI. Still True. (Lahtee)	own lineage, not independent	Institutional authority is derivative, not a source, of epistemic value — the same possession/constitution distinction H2 draws.
cite-own-zero-readout-h2-004	What a Zero Readout Certifies (Lahtee)	own lineage, not independent	Illustrates zero-institutional-apparatus rigour in a formal setting — weak indirect support only.

Table 4: H2 (conceptual proposition, this paper's own advanced claim) against 16 verified citation cards, 7 own-lineage. Diversity audit (`h2/litreview_manifest.yaml`): 3 Thai / 13 English, stance *agrees*:6 *disagrees*:2 *orthogonal*:7 *undetermined*:1. `concentration_flags: []`; `zero_disagree_flag: false` — this is the one proposition of the three whose manifest itself records a nonzero disagree count.

repo-level, not a paper-level, control: it did not need to run specifically against this paper's files this session (none of this paper's content matched a denylisted pattern), but it is the mechanical backstop this paper's own public-facing disclosures rely on, and its presence in CI — rather than only as a manual pre-publish step — is itself part of what this paper's demonstration reports: a leak gate that runs on every push, not only when someone remembers to ask for it.

Independent review of this paper's own claims

The three project claim cards behind this paper (Section A) were themselves put through the independence ladder before this draft was finalized: three cross-vendor AI review routes (R1 Falsifier, R2 HostileReviewer, R3 SourceAuditor; independence class I3), producing nine review reports in total (three routes × three cards, `projects/GLS-2026-001_rigour-without-infrastructure/reviews/*.review_report.json`), read each card independently of the session that drafted it. All nine converged on one defect class: statements written as results when the evidence behind them was one self-application run plus same-lineage or adjacent-domain citations, not a direct test of the mechanism stated. Concretely: H1's "can co-produce knowledge" was rewritten to "could, as a design proposition," with the paper's own n=1 self-application named explicitly as one demonstration readout, not a general test; H2's categorical "rigour is not conferred by institutions" was rewritten to a hedged conceptual proposition ("proposed here … at least in part"), with every same-lineage supporting citation now carrying an explicit not-independent note; H3's asserted "rises measurably" was rewritten to an untested hypothesis, "would rise … no measurement has been made, n=0." Every individual finding from all three routes is logged as a resolved dissent record on its originating claim card (`five_questions.tested.dissent_records`), not silently absorbed into a cleaner-looking rewrite. As of this draft, each card's `independent_check.status` is **PENDING**: checked by cross-vendor routes and revised accordingly, but not yet re-checked by a further independent route, and not yet approved by the founder for release (Section A). This section is offered as a working example of the independence ladder catching over-scope in this paper's own output — one instance, at I3, on one paper — not as proof that the ladder works in general; that broader claim would itself need the same discipline (H1's own falsifier, Section 5) before it could be asserted.

8 Limitations and disclaimers

D-STANDPOINT

See Section 2. This paper is written from a social-enterprise-practitioner/citizen standpoint, not disciplinary authority; a reader who skips to this section should read

Citation card	Source	Relation	Bearing
cite-selfpreference-judge-001	Wataoka, Takahashi & Ri 2024/25, Self-Preference Bias in LLM-as-a-Judge	cited	Weak, context-only: finds familiarity/self-preference bias, a mechanism relevant to why same-vendor re-checking may under-detect, but does not itself run a cross-vendor-vs-same-vendor comparison — cross-vendor review downgraded this from moderate to weak, scope narrowed to `CONTEXT_ONLY/SUPPORTS_GENERAL_MECHANISM`.
cite-crossmodel-codereview-002	Xiang et al. 2026, Cross-Model LLM Code Review, arXiv:2607.21656	different	The one source that actually runs H3's own comparison: the effect is asymmetric and direction-dependent (one direction helps a lot, the reverse hurts), and same-vendor re-review also improved one condition.
cite-toohconsistent-003	Tan et al. 2025, Too Consistent to Detect (EMNLP)	same	Weak, context-only: same-model repeated sampling cannot catch "self-consistent" errors, relevant to but not a direct test of H3's cross-vendor mechanism — cross-vendor review downgraded this from moderate to weak, scope narrowed to `CONTEXT_ONLY/SUPPORTS_GENERAL_MECHANISM`.
cite-multiagent-debate-004	Du et al. 2023, Multiagent Debate	cited	Multi-instance debate improves factuality without varying vendor identity — the active ingredient could be plurality, not cross-vendor difference.
cite-selfcorrect-reasoning-005	Huang et al. 2023/24, LLMs Cannot Self-Correct Reasoning Yet	cited	Tests zero-external-signal correction, a weaker condition than a same-vendor re-check with a fresh adversarial prompt — an imperfect analogy, flagged as such.
cite-legalai-india-006	John, Singh & Panachakel 2026, Can Legal AI Know When It Is Wrong?	cited	Three vendors' error rates differ nearly 5x on identical inputs, consistent with H3's premise, but the paper's own lever for improved catching is human experience, not cross-vendor AI review.
cite-bangladesh-readiness-007	Sultana et al. 2026, Bangladesh AI Readiness	cited	Names a capacity defeater of H3's practical reach (infrastructure to run any independent review at all), not of H3's empirical claim.
cite-iseas-sea-governance-008	Fong 2026, AI Governance Policies in Southeast Asia	cited	Institutional-capacity constraint on deploying cross-vendor review as policy, orthogonal to H3's mechanism claim.
cite-cofact-thai-detector-009	Cofact Thailand 2025, AI-image-detection reliability	cited	Adjacent domain (detecting whether content is AI-made, not whether AI-authored claims are honest) — genuinely different problem.
cite-mindstudio-crossvendor-010	MindStudio blog 2026, Cross-Vendor AI Agent Review	same	Weak, context-only: states H3's own mechanism in practitioner language, but with no measurement, sample, or method of its own — cross-vendor review scoped this `CONTEXT_ONLY/SUPPORTS_GENERAL_MECHANISM`, flagged as concentration risk, not equivalent-weight evidence.
cite-own-blackbox-log-h3-006	Blackbox Log (Lahtee)	own lineage, not independent	No claim about detection rates.
cite-own-knowledge-stabilized-translation-h3-007	Knowledge as Stabilized Translation (Lahtee)	own lineage, not independent	Stability under variation is the marker of knowing well — conceptually the same move H3 makes; does not specify vendor identity as the variation dimension.
cite-own-readout-condition-h3-002	The Readout Condition (Lahtee)	own lineage, not independent	Disclosure is structural, not optional; does not specify the auditor must be a different AI vendor.
cite-own-readout-genesis-h3-003	Readout Genesis Standalone Synthesis (Lahtee)	own lineage, not independent	Does not separate same-vendor from cross-vendor checking as a variable.
cite-own-standalone-scholar-h3-005	The Standalone Scholar (Lahtee)	own lineage, not independent	Independent validation in this paper's stronger sense requires human friction — cross-vendor AI (I3) is a weaker approximation.
cite-own-written-ai-h3-001	Written by AI. Still True. (Lahtee)	own lineage, not independent	Reinforces that a checker's identity/route matters to what its check can license — no detection-rate comparison.
cite-own-zero-readout-h3-004	What a Zero Readout Certifies (Lahtee)	own lineage, not independent	Machine-checked verification, where available, is a strictly stronger independence route than cross-vendor AI review.

Table 5: H3 (empirical proposition, explicitly untested) against 17 verified citation cards, 7 own-lineage. Diversity audit (`h3/litreview_manifest.yaml`): 1 Thai / 6 English, stance *agrees*:3 *orthogonal*:4 *undetermined*:10 by the manifest's own convention. `concentration_flags: []`; `zero_disagree_flag: false`.

every claim above as offered from that position and access level, never from institutional credential.

D-NONEXPERT
The author is not a substitute for a credentialed expert in citizen-science studies, philosophy of science, library/information science, or empirical AI-evaluation methodology — the disciplines this paper draws on without claiming standing in. Medicine, religious/fiqh authority, law, engineering, anthropology, political science, and formal computer science beyond what is mechanically executed here are explicitly not claimed.

D-SCOPE
Not applicable in the empirical-population sense (this is a CONCEPTUAL genre paper, Section 4). Where a source's own sample size or population matters to a comparison row (Tables 3 to 5), it is stated inline in that row rather than in one blanket population statement.

D-NONCLAIM
This paper does *not* claim: that H1's mechanism has been run as a study; that H2's conceptual-portability claim has been tested against real readers; that H3's seeded-error benchmark has been built or executed; that the 48 citation cards behind Tables 3 to 5 constitute a systematic or exhaustive review (`search_mode: TARGETED_SEARCH` / `SCOPING_SEARCH`, never `SYSTEMATIC_REVIEW`); that any finding here is a medical, legal, or religious determination; or that cross-vendor AI checking (I3) is equivalent to, or a substitute for, an I5 human review.

D-AIFILL
See Section 9. AI (the AI assistant, "the AI assistant" seat) drafted candidate hypotheses H1/H2/H3, the literature-review runs, the claim cards' `current_evidence`, `retrieved_tool_evidence`, `retained_record_route`, `model_calibration_assumption`, `prompt_system_constraint`, and `decision_policy` fields, and this paper's own prose and LaTeX build. None of this is treated as evidence on its own say-so (Section 9).

D-TIER
The single highest tier claimed anywhere in this paper is DR (declared bridge / human narrative synthesis over a specified mechanism), with one FINITE_DIAGNOSTIC

exception: the mechanically executed checks reported in Section 7 (schema validation, disclaimer emission, manifest gate reads) and this paper's own LaTeX compile. It cannot go higher because none of H1/H2/H3 has been run as a study, and no claim card behind this paper has passed a class-5 (I5) independent human check.

D-INDEPENDENCE

Every citation card behind Tables 3 to 5 was checked at independence class I3 (decorrelated cross-vendor AI route, `claim_match_verified_by: route:cross-vendor-ai`) for `claim_match`, and mechanically (I3-adjacent, Crossref/OpenAlex lookup) for `metadata_verified`. The three project claim cards themselves (Section A) have additionally been read by three cross-vendor AI review routes at I3 (Section 7) and revised in response, and now sit at `independent_check.status: PENDING` — checked and revised, but not yet re-checked by a route independent of that review pass, and not yet approved by the founder for release. No external or institutional validation is sought or treated as available as a lever that would make any claim here more true; an outside reader's input, if it comes, is optional-level evidence at whatever independence class it actually reaches, never a certification gate.

D-DVP-NOT-K2

A decorrelated-vendor AI review pass (DVP) is at most an I3 route. **DVP $\neq$ K2**: no combination of AI-only checks, however many vendors, raises this paper's claims to K2. This paper's K-state is `K0` (Section 8.1); no claim in it is asserted above K0/Dr.

D-LEGAL-NEQ-EPISTEMIC

None identified — no license, permit, or regulatory approval is referenced by this paper's claims. The general rule still holds: were one ever cited, it would make an action lawful, never raise epistemic warrant or certify a claim as true.

D-NOT-DIAGNOSTIC

Not applicable — this paper makes no health, clinical, or treatment recommendation of any kind.

D-TRANSLATION

Not applicable: per founder ruling (BBL-2026-09-04-099, Section B), this concept paper is published in English only, with no Thai edition. Every founder line quoted in Section B nonetheless stays Thai, verbatim, untranslated in place, with an English gloss alongside it — the English-only rule governs the paper's own prose, not the Blackbox Note's raw lines. Within the underlying claim cards, the Thai statement fields themselves carry `translation_status: machine` (machine translation, not human-reviewed).

D-DISSENT-PRESERVED

Nine dissent records are logged across the three claim cards (three cross-vendor routes × three cards, Section 7), each with a `resolved: true` entry naming what was accepted and how the card changed — none is deleted or silently folded into a cleaner rewrite; every one remains readable in the card's own `five_questions.tested.dissent_records` field. The literature-review dialogue tables (Tables 3 and 4) separately carry the closest real dissent this project has found against its own hypotheses (the Mayes 2022 boundary-work finding against H1 [cite-cstp-boundary-work-001]; the Scholarly Kitchen credentialism case against H2 [cite-rigour-without-infrastructure-004]); these are stated as open challenges in Section 6, not resolved or smoothed over.

D-COMPARISON

This paper does not claim priority or precedence. Section 6 states every source relation as *agrees*, *disagrees*, or *orthogonal/undetermined* — never as a priority contest, never as "no other work does X."

D-CITATION-UNVERIFIED

None identified for the 48 citation cards cited in Tables 3 to 5 — every one carries `status: VERIFIED` (`metadata_verified: true`, `claim_match_verified: true`) in its litreview manifest as of 2026-09-04; none is cited from memory (Section 7, BBL-2026-09-04-100/101). Where a source's own full text was not fetched (e.g. `fetch_status: PAYWALLED_ABSTRACT_ONLY`), the citation card's `scope` field names the resulting evidentiary weakness (`PARAPHRASE` rather than `DIRECT_QUOTATION`) inline in the relevant table row rather than silently upgrading it.

D-BLACKBOX-NOTE

This paper carries the mandatory Blackbox Note appendix (Section B). Note id: `BB-2026-09-04-01` (`projects/GLS-2026-001_rigour-without-infrastructure/bl` curated public source: `blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md`; public log entries `BBL-2026-09-04-083, -086, -088, -089, -091, -095, -097, -098, -099, -100, -101` (`blackbox/log/entries.jsonl`, concept DOI `10.5281/zenodo.22302518`).

D-K-STATE

Overall K-state: `K0` (internal working note). Every load-bearing claim in this paper — H1, H2, H3, and every finding in Section 7 — shares this same K0 state; there is no per-claim exception. This matches the three underlying project claim cards' own `k_state: K0` (Section A).

D-AUTHORSHIP

See Section 9. The problem, the research question, and the selection among H1/H2/H3 are the founder's own act (Section 2); AI drafted every candidate, citation search, and this paper's prose, disclosed per claim (`produced_by`, Table 6), and never substitutes for the author's judgment or responsibility (**AIContribution $\neq$ EpistemicResponsibility**).

D-REVISION-LIVE

The three underlying claim cards (`GLOSA-CC-20260904-0201/0202/0203`) are at revision 3 of an actively edited project, the second revision having been driven by the nine cross-vendor review findings described in Section 7; `status: Draft` and `independent_check.status: PENDING` on all three reflect that the founder has not yet signed off on any specific card instance, even though the founder has already ruled on genre (BBL-097) and hypothesis selection (BBL-095) at the project level.

D-NO-EPISTEMIC-VETO
No agency — including any AI vendor, reviewer, or institution named in this paper — holds an epistemic veto over H1, H2, or H3 merely by title, credential, or infrastructure ownership; any agency may propose, use, test, challenge, fork, translate, revise, restrict, or withdraw this work without needing permission, provided lineage is preserved (per the repo's own knowledge-validation stance, `AGENTS.md` (identical in the per-vendor gate files)).

D-CANDIDATE-STATUS
H1, H2, and H3 are candidate hypotheses drafted by AI; the act of selecting among them (all three, per BBL-2026-09-04-095) was the founder's alone (Section 2). No claim card here asserts candidate status was itself a form of testing.

D-EXTERNAL-INPUT
The lens this paper's readout-translation and hypothesis-drafting steps were performed under is named, cited, and linked throughout (Section 3); naming it credits its source and lets a reader replace it — it does not make the lens, or any hypothesis drafted under it, true.

8.1 Knowledge state

This paper's overall knowledge state is **K0** (internal working note): every load-bearing claim in it, including Section 7's findings, shares this K0 state. It is not K1 (public-provisional with a named independence class per claim and no class-5 human check yet) because the three underlying claim cards' own `status` field is still `Draft` and `independent_check.status: PENDING` (checked by three cross-vendor routes and revised, not yet re-checked by a further route or founder-approved, Section 7) — the founder has ruled on genre and selection at the project level but has not yet signed off on any specific card instance (Section A).

9 AI-ASSISTANCE DISCLOSURE

This paper is a human–AI co-production, not AI-only automation and not human-only authorship. A generative-AI assistant (vendor recorded in the local cooking log, not named here per gate rule 9) assisted with literature retrieval and card drafting, the three claim cards' field content, the conversation-with-the-literature tables, and this LaTeX build. **AI output is not treated as evidence.** Every claim above is accepted only when it matches a retained source record (a citation card's `exact_passage`, the Blackbox Log's own entries, a mechanically executed command's own output) or a machine-checkable result reported in Section 7. The human holds the standpoint, the falsifier judgment, verification of load-bearing citations at the founder's own discretion, and the final public commitment (**AIContribution $\neq$ EpistemicResponsibility**).

10 WHAT WOULD CHANGE OUR MIND

Per hypothesis, stated plainly rather than left implicit:

- **H1** would be abandoned if a genuine I5 reviewer, working from a schema-passing claim card, found a false answer at Q2 or Q3 — showing the mechanical check gives false reassurance rather than tracking the property it names. It would be strengthened, not proven, by a corpus comparison at I3+ (with a subset at I5) against a matched non-card corpus.
- **H2** would be abandoned if a reader study found that credentialed audiences (editors, grant reviewers, a practitioner's own community) still required an institutional signal even after being shown a complete, well-formed claim card — i.e. the credential doing load-bearing work the card's fields could not substitute for. The Scholarly Kitchen credentialism case (`cite-rigour-without-infrastructure-004`) is already the closest real instance of exactly this challenge and is not explained away.
- **H3** is untested; the one source that actually ran its comparison (`cite-crossmodel-codereview-002`) found a direction-dependent, asymmetric effect rather than a uniform cross-vendor-beats-same-vendor pattern. A seeded-error benchmark that reproduces that asymmetry at the claim-card Q3/Q4 grain would be evidence against H3 as a general, direction-agnostic claim; a benchmark that finds a uniform, direction-independent rise would be evidence for it.

11 CONCLUSION

This paper states three falsifiable propositions about whether a claim-card discipline can substitute for institutional infrastructure in making a standalone scholar's claim reader-checkable, places each against verified literature rather than memory or silence, and then runs the same discipline on its own construction — finding, honestly, a validator warning, a caught set of composite quotes that changed the method's own citation rule mid-project, a router/manifest mismatch the founder had to catch by hand, and a small stance-counting discrepancy in the project's own summary statistics. None of this is smoothed into a clean success story: every claim here stays DR, every underlying card stays K0 and unsigned, and H3 stays explicitly untested. What is offered is not a demonstrated result but a specified mechanism, an honest account of what happened when it was pointed at itself, and a named falsifier for each of the three ways it could be shown wrong.

A CLAIM MATRIX

Per Chair Ruling v1 §B2, this is a print-space compression of the three full claim cards (`projects/GLS-2026-001_rigour-without-infrastructure/clai`
field-presence here is mechanically checkable, correctness is what the independent check (Q5) exists for.

ID	Informal statement	Evidence / support	Tier	Produced by	Responsible
C1	H1: if every claim is recorded as a claim card tied to E-A-D and passed through the independence ladder before release, then a standalone scholar with no institutional affiliation *could, as a design proposition,* co-produce checkable-rigour knowledge with AI.	15 verified citation cards (Table 3); `GLOSA-CC-20260904-0201`	DR	joint (data→inference); human (inference→claim)	human
C2	H2: rigour is proposed here, as a conceptual contrast, to be *at least in part* a property a single claim can carry in its own recorded structure, not conferred solely from outside by institutional machinery.	16 verified citation cards (Table 4); `GLOSA-CC-20260904-0202`	DR	joint (data→inference); human (inference→claim)	human
C3	H3: cross-vendor (I3) checking of undisclosed AI-fill/assumptions *would rise* measurably compared to same-vendor (I0/I1) re-checking. **Untested; no seeded-error benchmark has been run by this project (n=0).**	17 verified citation cards (Table 5), all analogical, none a direct test of Q3/Q4 detection; `GLOSA-CC-20260904-0203`	DR	joint (data→inference); human (inference→claim)	human

Table 6: Claim Matrix. All three cards: `k_state: K0`, `status: Draft`, `independent_check.status: PENDING` (checked by cross-vendor routes R1–R3 and revised, Section 7; not yet re-checked by a further route or founder-approved), `independence_class: I3` (the citation-verification route, not the card's own sign-off).

A.1 Five-questions detail

B Blackbox Note: how this work was made

(Thai project term: "Bantuek Klong Dam".) Curated verbatim lines — typos kept, never polished or translated in place — followed by the cooking log. Full record archived separately (`projects/GLS-2026-001_rigour-without-infrastructure/blackbox_note.yaml`, note id `BB-2026-09-04-01`; public curated source `blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md`; public log entries cited by id below, `blackbox/log/entries.jsonl`, concept DOI `10.5281/zenodo.22302518`).

C Human Mastery Gate

Before submission the author confirms they can defend, without AI assistance: (1) the problem (Section 1) (2) the major literature families (Section 6) (3) the closest constructs (own-lineage rows, Section 6) (4) the precise inadequacy (the contaminated concept named in Section 1) (5) the proposed mechanism (H1, Section 5) (6) the boundary conditions ($E \wedge A \wedge D$, Section 3) (7) the strongest counterargument (the Scholarly Kitchen credentialism case, Table 4) (8) the strongest counterexample (the direction-dependent cross-model result against H3, Table 5) (9) the falsifying record for each proposition (Section 5) (10) the discriminating empirical test (H3's not-yet-run seeded-error benchmark). **Gate status: PASS WITH NAMED GAPS** — the founder has not yet performed the class-5 (I5) sign-off on any of the three underlying claim cards (Section A); this gap is named, not silently dropped.

D Provenance and lineage

Status vocabulary: PRESERVE_EXACT, PRESERVE_FUNCTION, EXPAND, SUPERSEDE. This paper PRESERVE_FUNCTIONs the claim-card schema and independence ladder from `design/FOUNDATION_v0.5.md` and EXPANDs it with a three-proposition literature conversation and a self-application demonstration specific to this project (GLS-2026-001). Direct ancestor: the Standalone Scholar dual-track architecture [cite-own-standalone-scholar-h1-005]. Core epistemic engine this paper's disclosure apparatus is built on: the Readout Condition [cite-own-readout-condition-h1-002].

References

cite-bangladesh-readiness-007. [cite-bangladesh-readiness-007] ARXIV: 2601.12934. citation card cite-bangladesh-readiness-007.yaml, records/lit/rigour-without-infrastructure/. URL `https://arxiv.org/abs/2601.12934`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Bangladesh, ranked 82nd in the 2023 Oxford Insights AI Readiness Index, exhibits significant deficits in technology capacity and research ecosystems, despite strong governmental visions... Findings reveal outdated curri...”.

cite-cofact-thai-detector-009. [cite-cofact-thai-detector-009] OFFICIAL_URL: https://blog.cofact.org/th/ai-detector-20052025/. citation card cite-cofact-thai-detector-009.yaml, records/lit/rigour-without-infrastructure/. URL `https://blog.cofact.org/th/ai-detector-20052025/`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “[Thai text – see citation card's own exact_passage field] (Artificial Intelligence: AI) [Thai text – see citation card's own exact_passage field]...”.

cite-crossmodel-codereview-002. [cite-crossmodel-codereview-002] ARXIV: 2607.21656. citation card cite-crossmodel-codereview-002.yaml, records/lit/rigour-without-infrastructure/. URL `https://arxiv.org/abs/2607.21656`. scope: DIRECT_QUOTATION; evidence_tier: fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation

ID	seen	ai_filled	assumed	tested (indep.)
C1	Blackbox Note L1; founder's abstract framing	candidate hypothesis text, evidence retrieval, card drafting (all disclosed, `disclaimers_emitted: D-AIFILL`)	no source tests the full combined H1 mechanism as one system	I3 (citation checks); card itself unsigned
C2	same L1; H2's own narrower conceptual framing	same AI-fill fields as C1	the two SUPPORTS sources are same-lineage; cannot alone move H2 past DR	I3; card itself unsigned
C3	same L1; H3's cross-vendor detection-rate framing	same AI-fill fields as C1	no seeded benchmark has been run; the closest analogous study found a direction-dependent effect	I3 (analogical citations only); card itself unsigned

Table 7: Five-questions detail keyed to Table 6. Full cards: `projects/GLS-2026-001_rigour-without-infrastructure/claims/GLOSA-CC-2026090` (and `-0202`, `-0203`).

ID	Kind	English gloss (AI assistant's translation, tier DR; effect on this paper)	Pointer
BBL-083	proposal	"the word 'observation' is abstract, it is a verb; if we started from the problem instead, is there a process like this? We propose focusing on the problem first." — separates "observation" (abstract verb) from "problem state" (concrete pre-existing thing); became FOUNDATION §2.1a (Section 1)	DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-083`
BBL-086/088	finding / correction	"if the problem is still ours, the question is still ours, and the selection of the hypothesis is still ours, and it leads to solving our problem, then use AI, use a cat, use a bear, go ahead — because that is the goal of knowledge: to solve a problem for someone." (088 corrects 086) — fixes which spine nodes must stay human: problem, question, and the *selection* of the hypothesis; became FOUNDATION §2.1c	DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-088` (corrects 086)
BBL-089	ruling	Ratifies the finer-grained spine diagram wording (AI-drafted, human-ratified) — became the spine diagram, FOUNDATION §2.1a	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-089`
BBL-091	instruction	Install and use the skill to build this article in full — became task B1/B2 launch	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-091`
BBL-095	ruling	"bring all three together" — selects all three hypotheses (H1, H2, H3); became `hypothesis_selection.yaml`	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-095`
BBL-097	ruling	"make this document an academic article, a concept paper" — sets genre to CONCEPTUAL, overriding the router's `systematic_review` output; became `genre_decision.yaml` (Section 4)	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-097`
BBL-098	instruction	Requires colliding own prior papers against world literature, not silence — became own-lineage rows, Section 6	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-098`
BBL-099	ruling	English only, no Thai edition; arXiv two-column theme, polished — became this document's language and theme	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-099`
BBL-100	ruling	"should we make it a hard rule: if a source is used it must carry a page and line locator plus a full link, never written from memory — that reduces the problem at the source, before the check." — mandates locator+link, source-first citation, after I3 caught composite quotes; became rule 17, §7.8 (Section 7)	DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-100`
BBL-101	instruction	"update it into the system" — adopts the citation rule immediately; became rule 17 adoption	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-101`

Table 8: Blackbox Note — every row is id / English gloss / pointer. Verbatim Thai lines are kept only in the cited records (the Blackbox Log, DOI `10.5281/zenodo.22302518`, or `blackbox/log/entries.jsonl`), by founder ruling `BBL-2026-09-04-103` (English-only paper); the glosses shown here are the AI assistant's translations, not the founder's own words, tier DR.

card status: VERIFIED; exact passage: "Claude review raises Codex drafts from 71.6% to 89.7% ($p\_BH = .001$); Codex self review raises them to 84.5% ($p\_BH = .022$). The reverse direction does not pay off: Codex reviewing Claude drafts drops the pass rate fro...".

cite-cstp-boundary-work-001. [cite-cstp-boundary-work-001] DOI: 10.5334/cstp.519. citation card cite-cstp-boundary-work-001.yaml, records/lit/rigour-without-infrastructure/. URL `https://doi.org/10.5334/cstp.519`. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: ""the legitimacy of citizen science is powerfully shaped by the perspectives of professional or credentialed experts" – in a 2010-2019 sample of 51 news articles, "perspectives of professional and institutionally affilia...".

cite-disclosure-not-documentation-001. [cite-disclosure-not-documentation-001] DOI: 10.1186/s41073-026-00245-8. citation card cite-disclosure-not-documentation-001.yaml, records/lit/rigour-without-infrastructure/. URL `https://doi.org/10.1186/s41073-026-00245-8`. scope: PARAPHRASE; evidence_tier: Dr; independence_class: I3; fetch_status: PAYWALLED_ABSTRACT_ONLY; citation card status: VERIFIED; exact passage: ""disclosure alone rarely clarifies how AI-assisted work was performed, what information was entered, how outputs were evaluated, or how human responsibility was maintained" – creating a gap between AI-use disclosure as ...".

cite-iseas-sea-governance-008. [cite-iseas-sea-governance-008] OFFICIAL_URL: https://www.iseas.edu.sg/articles-commentaries/iseas-perspective/2025-13-what-is-shaping-artificial-intelligence-ai-governance-policies-in-southeast-asia-by-kristina-fong/. citation card cite-iseas-sea-governance-008.yaml, records/lit/rigour-without-infrastructure/. URL `https://www.iseas.edu.sg/articles-commentaries/iseas-perspective/2025-13-what-is-shaping-artificial-intelligence-ai-governance-policies-in-southeast-asia-by-kristina-fong/`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status:

Step	By	Input lines	Output	What changed
lens_in	ai-assistant	L1	`lens_translation.md`	Drafted `question_readout`, `local_contrast_space_X`, `access_relation_R`, `claim_function_Phi_z0` from L1 under the Readout Universe lens; not yet independently reviewed.
lens_out	ai-assistant	L1–L13	`hypotheses.md`	Drafted H1/H2/H3 as candidate world-language hypotheses; selection reserved to and performed by the founder (BBL-095).
lit_review	ai-assistant	H1/H2/H3	48 citation cards, 3 `litreview_manifest.yaml`	Ran targeted/scoping searches per hypothesis (not systematic), froze manifests, computed diversity/accuracy audits.
claim_card	ai-assistant	hypotheses + literature	`GLOSA-CC-20260904-0201/0202/0203`	Filled full-shape cards; validated `PASS_WITH_LIMITS`; not yet founder-signed.
paper_build	ai-assistant	all of the above	this document	Drafted the concept paper's prose, tables, and LaTeX build; disclosed throughout, never authored.

Table 9: Cooking log (Thai project term: “karn pung”) — append-only.

FETCHED; citation card status: VERIFIED; exact passage: “"The capabilities of DPAs across Southeast Asia are very diverse. As such, the use of DPAs as central market surveillance authorities may be viable for some, but not others... existing DPA capacities will need to be bols...”.

cite-legalai-india-006. [cite-legalai-india-006] ARXIV: 2608.21089. citation card cite-legalai-india-006.yaml, records/lit/rigour-without-infrastructure/. URL `https://arxiv.org/abs/2608.21089`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"students reporting multiple prior encounters with fabricated citations reported a higher descriptive mean verification score (4.2/5) than students reporting no such encounters (2.8/5)."”.

cite-llm-hallucination-survey-001. [cite-llm-hallucination-survey-001] ARXIV: 2510.06265. citation card cite-llm-hallucination-survey-001.yaml, records/lit/rigour-without-infrastructure/. URL `https://arxiv.org/abs/2510.06265`. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Hallucinations undermine the reliability and trustworthiness of LLMs, especially in domains requiring factual accuracy." Self-consistency detection methods "cannot catch hallucinations that are confidently and consisten...”.

cite-mindstudio-crossvendor-010. [cite-mindstudio-crossvendor-010] OFFICIAL_URL: https://www.mindstudio.ai/blog/cross-vendor-ai-agent-review-claude-codex. citation card cite-mindstudio-crossvendor-010.yaml, records/lit/rigour-without-infrastructure/. URL `https://www.mindstudio.ai/blog/cross-vendor-ai-agent-review-claude-codex`. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Open; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"OpenAI’s Codex and Anthropic’s Claude don’t share training data, architectures, or fine-tuning pipelines. When Codex tends to miss a particular class of security issue...Claude often catches it, and vice versa." ... "Wh...”.

cite-multiagent-debate-004. [cite-multiagent-debate-004] ARXIV: 2305.14325. citation card cite-multiagent-debate-004.yaml, records/lit/rigour-without-infrastructure/. URL `https://arxiv.org/abs/2305.14325`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We present a complementary approach to improve language responses where multiple language model instances propose and debate their individual responses and reasoning processes over multiple rounds to arrive at a common ...”.

cite-ocsdnet-honf-001. [cite-ocsdnet-honf-001] OFFICIAL_URL: https://ocsdnet.org/projects/hita-ordo-natural-fiber-honf-foundation/. citation card cite-ocsdnet-honf-001.yaml, records/lit/rigour-without-infrastructure/. URL `https://ocsdnet.org/projects/hita-ordo-natural-fiber-honf-foundation/`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “The main objective is to understand how open science in Indonesia, Thailand and Nepal operates between formal and informal institutions of knowledge and technology transfer and the dynamic of North/South/South networks o...”.

cite-own-blackbox-log-h1-006. [cite-own-blackbox-log-h1-006] DOI: 10.5281/zenodo.22302518. citation card cite-own-blackbox-log-h1-006.yaml, records/lit/rigour-without-infrastructure/. URL `https://doi.org/10.5281/zenodo.22302518`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Blackbox Log – a daily, append-only, verbatim record of the author’s raw voice (one pre-release correction is disclosed in BBL-2026-09-04-082), findings and questions (Thai is the source of truth)... Contains the author...”.

cite-own-blackbox-log-h2-006. [cite-own-blackbox-log-h2-006] DOI:

10.5281/zenodo.22302518. citation card cite-own-blackbox-log-h2-006.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22302518. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Blackbox Log – a daily, append-only, verbatim record of the author’s raw voice (one pre-release correction is disclosed in BBL-2026-09-04-082), findings and questions (Thai is the source of truth)... Contains the author...”.

cite-own-blackbox-log-h3-006. [cite-own-blackbox-log-h3-006] DOI: 10.5281/zenodo.22302518. citation card cite-own-blackbox-log-h3-006.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22302518. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Blackbox Log – a daily, append-only, verbatim record of the author’s raw voice (one pre-release correction is disclosed in BBL-2026-09-04-082), findings and questions (Thai is the source of truth)... Contains the author...”.

cite-own-civilization-of-knowledge-h2-007. [cite-own-civilization-of-knowledge-h2-007] DOI: 10.5281/zenodo.18943971. citation card cite-own-civilization-of-knowledge-h2-007.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.18943971. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"What changes across civilizations is not only the accumulation of truths, but the succession of agencies authorized to interpret the world’s signals and stabilize those interpretations as social reality... These do not ...”.

cite-own-knowledge-stabilized-translation-h3-007. [cite-own-knowledge-stabilized-translation-h3-007] DOI: 10.5281/zenodo.18925129. citation card cite-own-knowledge-stabilized-translation-h3-007.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.18925129. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"knowledge is best understood as stabilized translation under bounded observation: a representation that remains stable across admissible variation, calibration, and cross-checking. The result is a framework that preserv...”.

cite-own-readout-condition-h1-002. [cite-own-readout-condition-h1-002] DOI: 10.5281/zenodo.22301318. citation card cite-own-readout-condition-h1-002.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301318. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"This paper develops the Readout Condition as a necessary architecture for source-relative epistemic warrant. Its core is a three-part norm: existence (every claim-level distinction has an epistemic provenance path), att...”.

cite-own-readout-condition-h2-002. [cite-own-readout-condition-h2-002] DOI: 10.5281/zenodo.22301318. citation card cite-own-readout-condition-h2-002.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301318. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"This paper develops the Readout Condition as a necessary architecture for source-relative epistemic warrant. Its core is a three-part norm: existence (every claim-level distinction has an epistemic provenance path), att...”.

cite-own-readout-condition-h3-002. [cite-own-readout-condition-h3-002] DOI: 10.5281/zenodo.22301318. citation card cite-own-readout-condition-h3-002.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301318. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"This paper develops the Readout Condition as a necessary architecture for source-relative epistemic warrant. Its core is a three-part norm: existence (every claim-level distinction has an epistemic provenance path), att...”.

cite-own-readout-genesis-h1-003. [cite-own-readout-genesis-h1-003] DOI: 10.5281/zenodo.21529456. citation card cite-own-readout-genesis-h1-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21529456. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"the paper inserts a typed belief layer between memory and claim evaluation, defines belief as a multidimensional agency-claim relation, and proposes that knowledge be attributed as an auditable status of a claim rather ...”.

cite-own-readout-genesis-h2-003. [cite-own-readout-genesis-h2-003] DOI: 10.5281/zenodo.21529456. citation card cite-own-readout-genesis-h2-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21529456. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage:

“"the paper inserts a typed belief layer between memory and claim evaluation, defines belief as a multidimensional agency-claim relation, and proposes that knowledge be attributed as an auditable status of a claim rather ...”.

cite-own-readout-genesis-h3-003. [cite-own-readout-genesis-h3-003] DOI: 10.5281/zenodo.21529456. citation card cite-own-readout-genesis-h3-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21529456. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"the paper inserts a typed belief layer between memory and claim evaluation, defines belief as a multidimensional agency-claim relation, and proposes that knowledge be attributed as an auditable status of a claim rather ...”.

cite-own-standalone-scholar-h1-005. [cite-own-standalone-scholar-h1-005] DOI: 10.5281/zenodo.22163849. citation card cite-own-standalone-scholar-h1-005.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22163849. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"It develops a dual-track architecture in which AI-assisted formation, open preprint staging, provenance controls, and machine-constrained checks generate defensible provisional assets, while independent human friction, ...”.

cite-own-standalone-scholar-h2-005. [cite-own-standalone-scholar-h2-005] DOI: 10.5281/zenodo.22163849. citation card cite-own-standalone-scholar-h2-005.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22163849. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"It develops a dual-track architecture in which AI-assisted formation, open preprint staging, provenance controls, and machine-constrained checks generate defensible provisional assets, while independent human friction, ...”.

cite-own-standalone-scholar-h3-005. [cite-own-standalone-scholar-h3-005] DOI: 10.5281/zenodo.22163849. citation card cite-own-standalone-scholar-h3-005.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22163849. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"It develops a dual-track architecture in which AI-assisted formation, open preprint staging, provenance controls, and machine-constrained checks generate defensible provisional assets, while independent human friction, ...”.

cite-own-written-ai-h1-001. [cite-own-written-ai-h1-001] DOI: 10.5281/zenodo.22301202. citation card cite-own-written-ai-h1-001.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301202. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Institutions remain indispensable, but their authority is derivative from epistemic work, not magical pedigree. AI is philosophically useful because it removes the human face from a claim and reveals how much epistemic ...”.

cite-own-written-ai-h2-001. [cite-own-written-ai-h2-001] DOI: 10.5281/zenodo.22301202. citation card cite-own-written-ai-h2-001.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301202. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Institutions remain indispensable, but their authority is derivative from epistemic work, not magical pedigree. AI is philosophically useful because it removes the human face from a claim and reveals how much epistemic ...”.

cite-own-written-ai-h3-001. [cite-own-written-ai-h3-001] DOI: 10.5281/zenodo.22301202. citation card cite-own-written-ai-h3-001.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301202. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Institutions remain indispensable, but their authority is derivative from epistemic work, not magical pedigree. AI is philosophically useful because it removes the human face from a claim and reveals how much epistemic ...”.

cite-own-zero-readout-h1-004. [cite-own-zero-readout-h1-004] DOI: 10.5281/zenodo.21665100. citation card cite-own-zero-readout-h1-004.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21665100. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We show that a zero readout is not an absence but a classification, and prove which one... The characterisation is machine-checked over the rationals without classical axioms."”.

cite-own-zero-readout-h2-004. [cite-own-zero-readout-h2-004] DOI: 10.5281/zenodo.21665100. citation card cite-own-zero-readout-h2-004.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21665100. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We show that a zero readout is not an absence but a classification, and prove which one... The characterisation is machine-checked over the rationals without classical axioms."”.

cite-own-zero-readout-h3-004. [cite-own-zero-readout-h3-004] DOI: 10.5281/zenodo.21665100. citation card cite-own-zero-readout-h3-004.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21665100. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We show that a zero readout is not an absence but a classification, and prove which one... The characterisation is machine-checked over the rationals without classical axioms."”.

cite-pgph-citizen-science-sea-001. [cite-pgph-citizen-science-sea-001] DOI: 10.1371/journal.pgph.0003696. citation card cite-pgph-citizen-science-sea-001.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC12370037/fullTextXML. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “Current study showed that people being the recipients of policies and programs, ready to have their voices included.”.

cite-rigour-without-infrastructure-001. [cite-rigour-without-infrastructure-001] OFFICIAL_URL: https://en.wikipedia.org/wiki/Open-notebook_science. citation card cite-rigour-without-infrastructure-001.yaml, records/lit/rigour-without-infrastructure/. URL https://en.wikipedia.org/wiki/Open-notebook_science. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: definition; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Open-notebook science is the practice of making the entire primary record of a research project publicly available online as it is recorded." (article also notes open-notebook projects have been published in peer-review...”.

cite-rigour-without-infrastructure-002. [cite-rigour-without-infrastructure-002] DOI: 10.5334/cstp.207. citation card cite-rigour-without-infrastructure-002.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5334/cstp.207. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"the field lacks an overall, widely accepted approach to research integrity, which would at least include training citizen scientists in concepts and methods of research integrity, establishing mechanisms to protect the ...”.

cite-rigour-without-infrastructure-003. [cite-rigour-without-infrastructure-003] DOI: 10.5334/cstp.6. citation card cite-rigour-without-infrastructure-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5334/cstp.6. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We surveyed the credibility-building strategies of 30 citizen science programs that monitor environmental aspects of the California coast. We identified a total of twelve strategies: Three that are applied during traini...”.

cite-rigour-without-infrastructure-004. [cite-rigour-without-infrastructure-004] OFFICIAL_URL: https://scholarlykitchen.sspnet.org/2026/08/14/guest-post-the-guild-and-the-gap-what-one-rejection-revealed-about-the-cost-of-credentialism/. citation card cite-rigour-without-infrastructure-004.yaml, records/lit/rigour-without-infrastructure/. URL https://scholarlykitchen.sspnet.org/2026/08/14/guest-post-the-guild-and-the-gap-what-one-rejection-revealed-about-the-cost-of-credentialism/. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “Preparing manuscripts takes training, practice and experience. If someone has no experience, no affiliation, no supervision or advisory colleagues, etc., then the system should be very careful with that person, especiall...”.

cite-rigour-without-infrastructure-005. [cite-rigour-without-infrastructure-005] DOI: 10.1093/joc/jqab027. citation card cite-rigour-without-infrastructure-005.yaml, records/lit/rigour-without-infrastructure/. URL https://api.crossref.org/works/10.1093/joc/jqab027. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “These decolonizing practices foreground data sovereignty, community ownership, and public ownership of knowledge resources as the bases of resistance to the colonial-capitalist interests of hegemonic Open Science.”.

cite-rigour-without-infrastructure-006. [cite-rigour-without-infrastructure-006] DOI: 10.3389/frma.2022.855198. citation card cite-rigour-without-infrastructure-006.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC9051436/fullTextXML. scope:

CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"The advancement of scientific research and raising the next-generation scientists in Africa depend largely on science access. The COVID-19 pandemic has caused discussions around open science (OS) to reemerge globally, e...”.

cite-rigour-without-infrastructure-007. [cite-rigour-without-infrastructure-007] DOI: 10.1371/journal.pgph.0003696. citation card cite-rigour-without-infrastructure-007.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC12370037/fullTextXML. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Citizen science (CS), a collaboration between people and scientists, is a viable approach utilizing citizens experiences in COVID-19 pandemic to manage future response. This study aimed to understand concepts, experienc...”.

cite-rigour-without-infrastructure-008. [cite-rigour-without-infrastructure-008] OFFICIAL_URL: https://so05.tci-thaijo.org/index.php/JTPLC/article/view/283964. citation card cite-rigour-without-infrastructure-008.yaml, records/lit/rigour-without-infrastructure/. URL https://so05.tci-thaijo.org/index.php/JTPLC/article/view/283964. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “Confirmability emphasizes neutrality, ensuring that interpretations are derived from actual data rather than the researcher’s imagination, often documented through an audit trail.”.

cite-rigour-without-infrastructure-009. [cite-rigour-without-infrastructure-009] OFFICIAL_URL: https://thestandard.co/thailand-open-science-unlock-ai-big-data-platform/. citation card cite-rigour-without-infrastructure-009.yaml, records/lit/rigour-without-infrastructure/. URL https://thestandard.co/thailand-open-science-unlock-ai-big-data-platform/. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"[Thai text – see citation card’s own exact_passage field] Ï[Thai text – see citation card’s own exact_passage field]Ï[Thai text – see citation card’s own exact_passage field]"”.

cite-rsos-citizen-epistemology-001. [cite-rsos-citizen-epistemology-001] DOI: 10.1098/rsos.231100. citation card cite-rsos-citizen-epistemology-001.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC10646465/fullTextXML. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “Thus, multiple perspectives enhance robust insight, and a multiplicity of perspectives is what democratic citizen science provides.”.

cite-rsos-preprint-credibility-001. [cite-rsos-preprint-credibility-001] DOI: 10.1098/rsos.201520. citation card cite-rsos-preprint-credibility-001.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC7657885/fullTextXML. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “In general, information related to field perceptions/usage of preprints was rated as less important by respondents, and information related to preprint authors (e.g. institutional information, previous work) received mor...”.

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fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Our findings reveal that LLMs assign significantly higher evaluations to outputs with lower perplexity than human evaluators, regardless of whether the outputs were self-generated. This suggests that the essence of the ...”.

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cite-toohconsistent-003. [cite-toohconsistent-003] DOI: 10.18653/v1/2025.emnlp-main.238. citation card cite-toohc onsistent-003.yaml, records/lit/rigour-without-infrastructure/. URL `https://doi.org/10.18653/v1/2025.emnlp-main.238`. scope: DIRECT_QUOTATION; evidence_tier: fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Motivated by the observation that self-consistent errors often differ across LLMs, we propose a simple but effective cross-model probe method that fuses hidden state evidence from an external verifier LLM. Our method si...”.